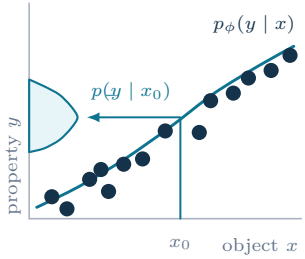
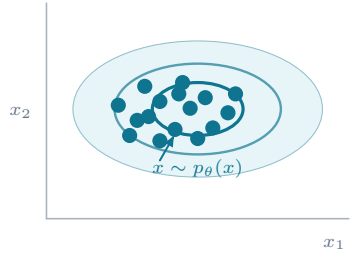
**(a) Discriminate**
**learn $p(y \mid x)$**



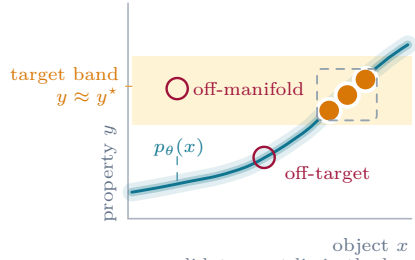

given an object $x_0$, read a whole property distribution off the fitted curve

**(b) Generate**
**learn $p_\theta(x)$**




density contours over object space; no property axis at all

**(c) Inverse design**
**sample $x \sim p(x \mid y^\star)$**




a candidate must lie in the band *and* on the manifold $p_\theta(x)$; each rose ring fails one test, the orange samples pass both